Sketching Polar Curves

Key Points

A relation between θ and r represents a curve. Remember

$r \geq 0$

We can calculate a number of points on the curve to get an indication of the shape.

The equation $r = a$ represents a circle centred at the origin with radius a.

The equation $\theta = \alpha$ is a half line from the origin making an angle α with the positive x-axis.

Basic Skills

Q1. Sketch the following curves

a) $r = 1$

b) $r = 5$

c) $\theta = 0$

d) $\theta = \frac{3\pi}{4}$

Q2. Evaluate the polar equation $r = 2\sin(3\theta)$ to find r at

a) $\theta = 0$ **b)** $\theta = \frac{\pi}{6}$ **c)** $\theta = -\frac{\pi}{4}$

Competency Skills

Q3. A curve has polar equation $r = 3\cos(\theta)$ for $0 \leq \theta < 2\pi$

a) Complete the table

b) Hence, sketch the curve

c) What can you say about the symmetry of the curve

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	$\frac{11\pi}{6}$
r								

Q4. Sketch curve has polar equation $r = \cos(2\theta) + \sin(\theta)$ for $0 \leq \theta < 2\pi$

Q5. Sketch curve has polar equation $r = \sin(\theta) + \theta$

a) For $0 \leq \theta < 2\pi$

b) For $-\pi \leq \theta < \pi$

Advanced Skills

Q6. A curve has equation $r = 2 - 4\sin(\theta)$ for $0 \leq \theta < 2\pi$

a) Without sketching the curve can you give a range of values of θ which will not be plotted

b) Without sketching the curve can you explain why it will be symmetric about the y-axis

Q7. Plot $r = \theta$ for all $\theta \geq 0$

Q8. A curve has polar equation $r = \tan\theta$ for $-\pi < \theta \leq \pi$

a) Explain what values of θ cannot be drawn and what happens to the radius as you approach them

b) Sketch the graph

Extension

Q9. Can you find a general formula for the polar equation of a circle with centre (a, b) and radius c.

Hint: Start with the Cartesian equation of the circle.